


$$F = C - P + 2$$

$\downarrow$ phases
 $\downarrow$ components

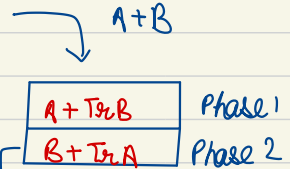
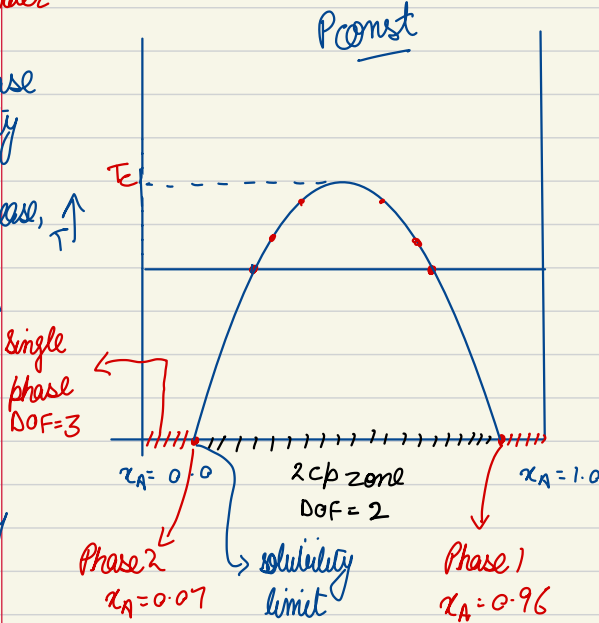
Liquid liquid extraction

2 immiscible liquids $\begin{pmatrix} A & x_A \\ B & x_B \end{pmatrix}$ $x_A + x_B = 1.0$

ex coffee & water

Temp increase
 solubility
 of A in B
 will increase, $\uparrow T$
 similarly
 for B in A

Dome of
 insolubility



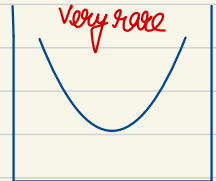
In reality they won't
 be completely
 immiscible so traces
 of A & B

Tie line $\rightarrow$ A line connecting
 two equilibrium
 pts

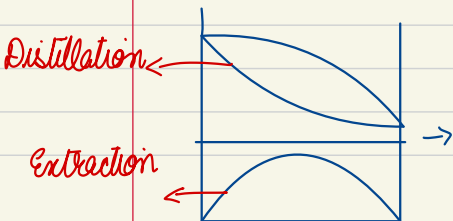
connects vapour & liquid
 composition at equilibrium

UCST system

$T_c \rightarrow$ upper critical solution temperature
 Above T_c A & B are completely miscible



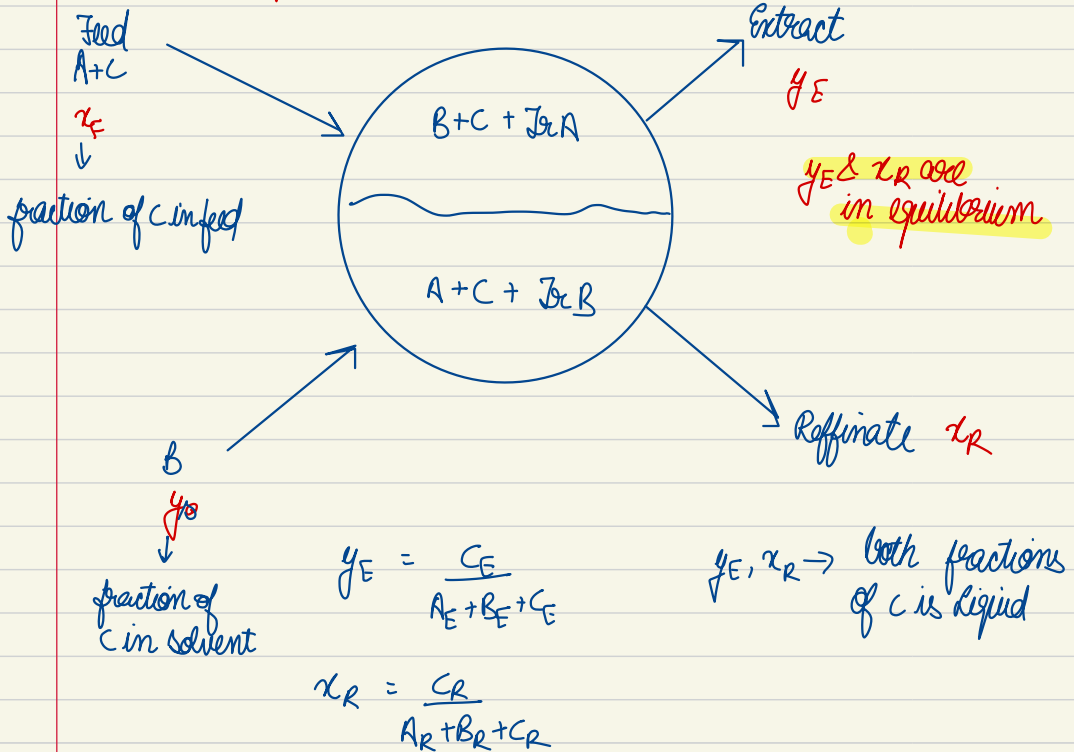
LCST
 Lower critical solution
 temp



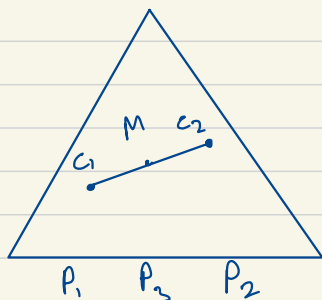
Continue heating after
 T_c , liquids will
 separate based on diff. in boiling pts

A, B immiscible
 C $\rightarrow$ miscible in both

$x, y \rightarrow$ both are fraction of C



30/8



$$C_1 = x_1 \quad l_1 \text{ kg}$$

$$C_2 = x_2 \quad l_2 \text{ kg}$$

$$l_1 + l_2 = M$$

$$l_1 x_1 + l_2 x_2 = M x_M$$

$$M \cdot \overline{MP_3} = l_1 \overline{C_1 P_1} + l_2 \overline{C_2 P_2}$$

$$\Rightarrow (l_1 + l_2) \overline{MP_3}$$

$$\frac{l_1}{l_2} = \overline{MP_3} - \overline{C_2 P_2}$$

Solubility $\rightarrow$ affinity for solvent molecules $\rightarrow \checkmark \checkmark$

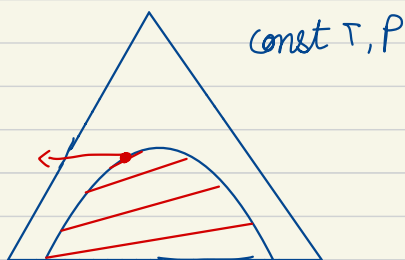
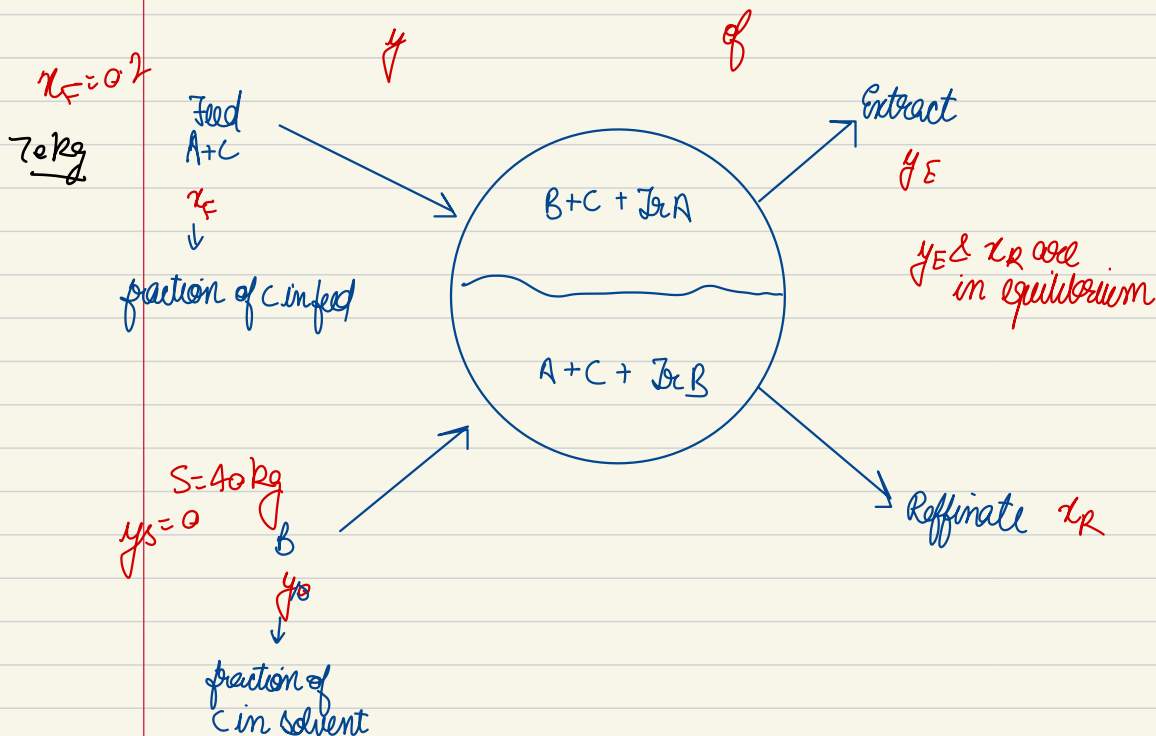


Plate
point



as conc. of C increases, zone of immiscibility decreases and coincides at plate point



$$M = F + S = 110 \text{ kg}$$

C balance

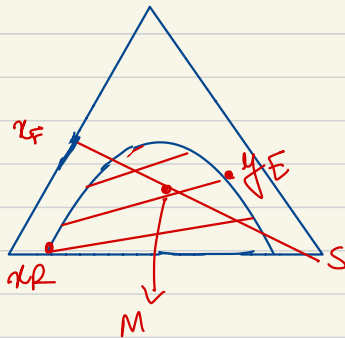
$$M x_M = F x_F + S y_S \rightarrow C$$

$$70 \times 0.2 = 110 \times x_M$$

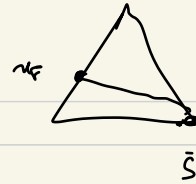
$$x_M = \frac{14}{110}$$

$$x_M = 0.127$$

$\hookrightarrow$ fraction of
 C in final
 mixture



$\Rightarrow$



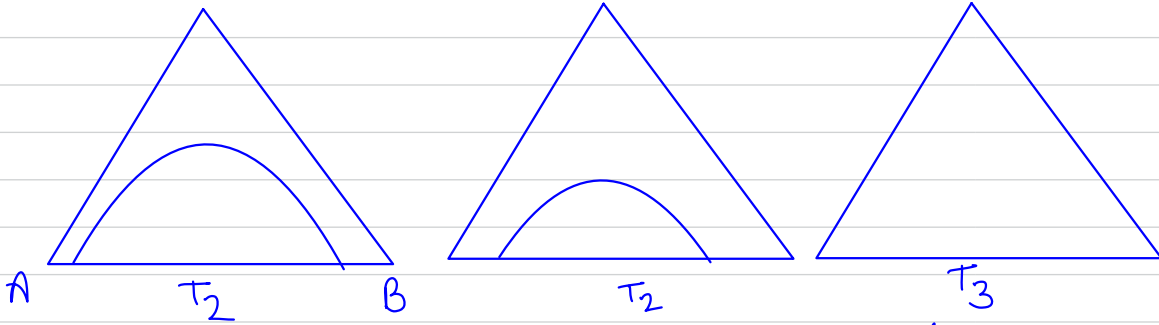
x_m is going to split into x_R & y_E

$$M = R + E$$

$$M x_m = R x_R + E y_E$$

we can find R & E

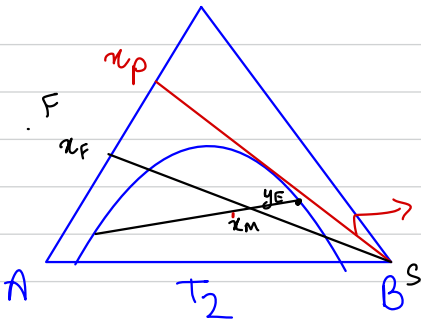
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Extraction is favoured at low temp

This graph shows
Above critical temp,
A & B are miscible

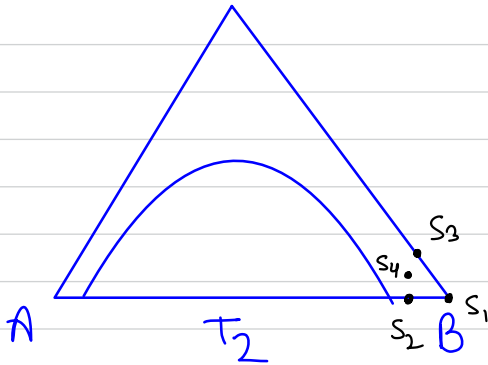
Extraction $\rightarrow$ temp should be/w bubble point & dew point



$$x_F < x_p$$

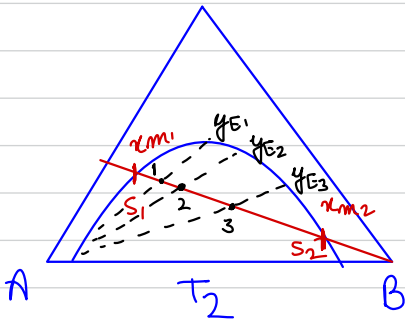
If $x_F > x_p$,
there can't
be separation
a homogenous
mixture would
be formed.

We want our x_m to lie in our
dome of immiscibility



If we draw tangents from S_1, S_2, S_3, S_4 we get different x_p i.e. extraction limit and we get maximum limit of x_p in case of S_2 and minimum in case of S_3 [cont. animation by A]

Whether x_m would lie in dome of immiscibility depends on relative position of F & S



$F, x_F, S, y_S \rightarrow \text{known}$

$M, x_M \rightarrow \text{unknown}$

Relative values of feed & solvent decide the x_m

For effective separation we want

x_m to lie b/w x_{m1} & x_{m2}

or b/w the dome of immiscibility

$F, x_F, y_S \rightarrow \text{known}$
what is allowable
max & min solvent
amount?

$S_1 \rightarrow \text{Min amount of solvent}$

$S_2 \rightarrow \text{Max amount of solvent}$

So ques arises why don't we use min. solvent amount?
→ We would get more y_E i.e. if we choose our solvent amount from ①, then $y_{E1} > y_{E2} > y_{E3}$

We get more quality extract, isn't it a win-win situation?

If we choose min/less solvent amount, our E would also be less i.e. we would get low quantity of good quality extract.

So, it is a trade off between amount of extract and quality of extract.

[Higher y_E , gives lower E]

Choice of Solvent

1) Selectivity :-

$$B = \frac{(\text{wt fraction of C in E}) / (\text{wt fraction of A in E})}{(\text{wt fraction of C in R}) / (\text{wt fraction of A in R})}$$

$$= \frac{\left(\frac{y_E}{x_R} \right) \times (\text{wt fraction of A in R})}{(\text{wt fraction of A in E})}$$

↓

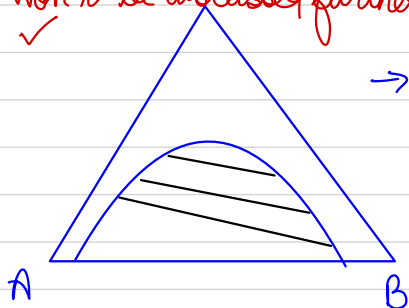
Partition coefficient

So necessary condition for effective separation

$$B > 1$$

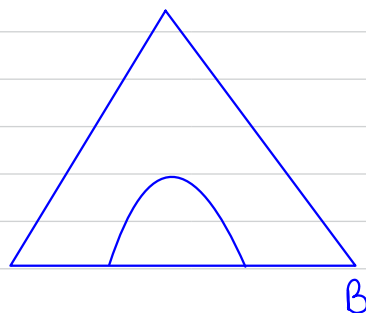
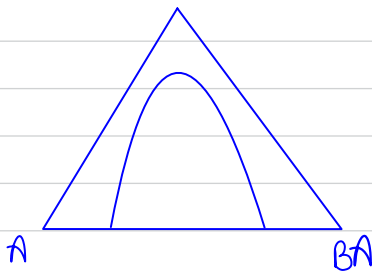
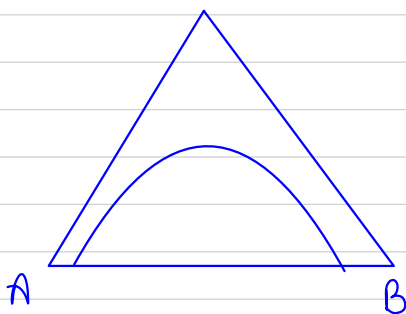
For $B=1$, the tie line would become horizontal [$y_E = x_E$]

won't be discussed further
✓



→ what if we take solvent B such that solubility of C in B is less than solubility of C in A then we get these type of tie lines

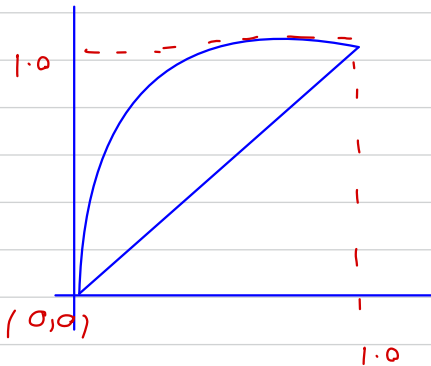
→ This can be done, when we have another method in which we can extract C from B very effectively



2. Partition coefficient
3. Degree of insolubility
4. Recyclability of the solvent
5. Density $\rightarrow$ density difference for stratification
6. Surface tension

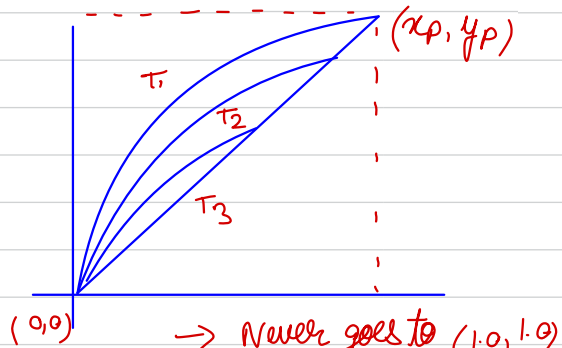
Two edges of every tie line is (x, y)

Plait point $\rightarrow$ where composition of c in extract & raffinate are same



For distillation

at different temperatures



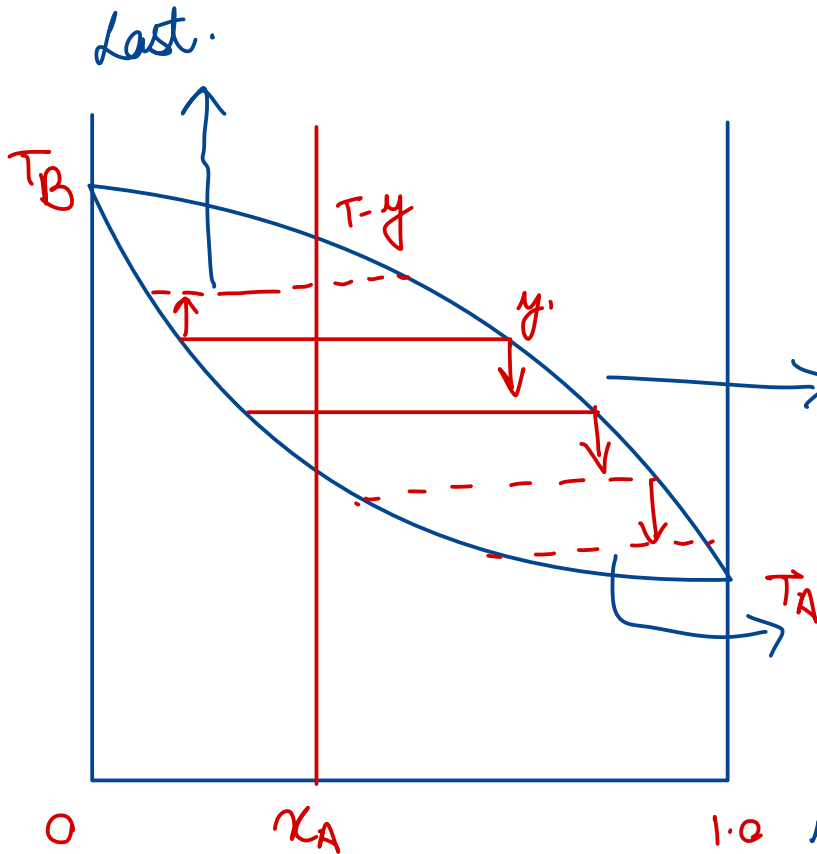
$\rightarrow$ Never goes to (1,0, 1,0)
like distillation

$\rightarrow$ Isothermal plot
For extraction

$$T_3 > T_2 > T_1$$

1/9/22

Distillation temp $\rightarrow$ should be bubble pt & dew pt of that composition



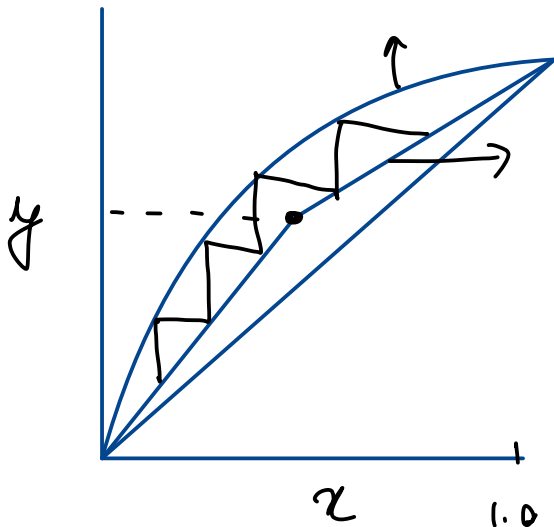
McCabe Thiele??

Each of these tie line is a stage

First stage $\rightarrow$ Top

by Top is more volatile material
which has lower BP.

Equilibrium plot

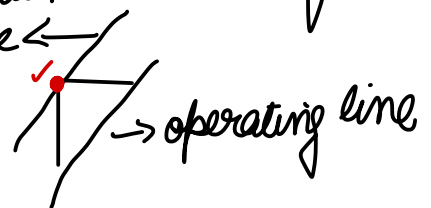


operating line

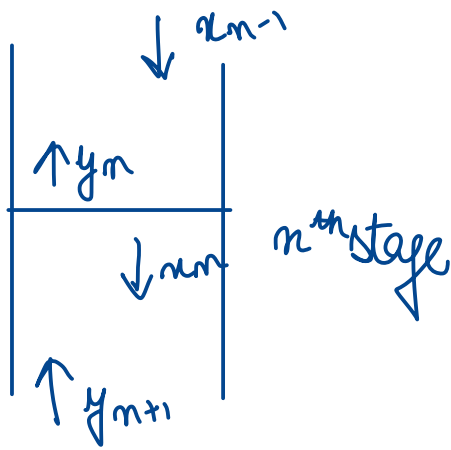
$\rightarrow$ what we think

Each step corresponds to one stage

Equilibrium curve



what everyone says is that each step is a stage but in reality only the point (where horizontal & vertical meet) is the actual stage



[Locus of all possible combinations
..... ?]

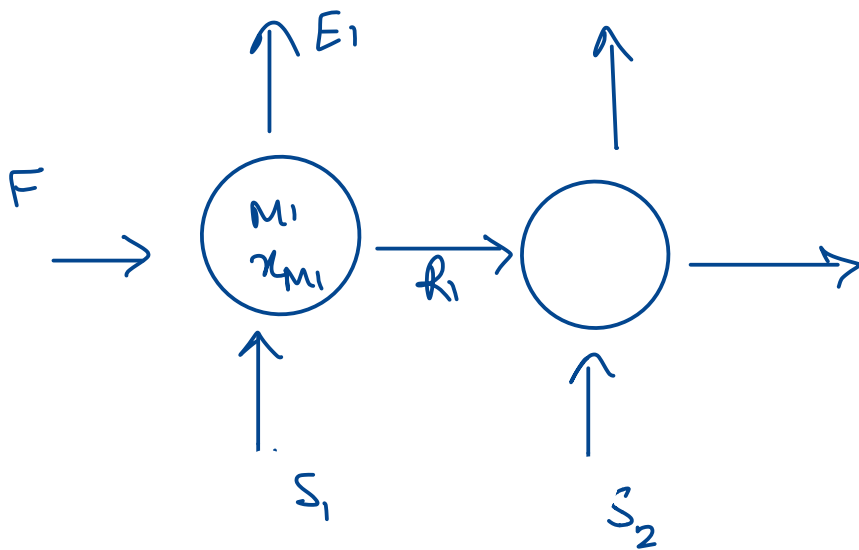
↳ operating line

$$y_{n+1} = \frac{R}{R+1} x_n + \dots$$

↳ eqn of operating line

more volatile compound $\rightarrow$ lower BP, higher vapour pressure

objective $\rightarrow$ How to extract more C from Raffinate??



F, S_1
 x_F, y_S

↓

y_E, x_R
 E_1, R_1

For second stage

$$R_1 + S_2 = M_2$$

$$R_1 x_{R_1} + S_2 y_S = M_2 x_{M_2}$$

